

ideals : prime ideal, maximal ideal.



We have seen subgroups and subrings

Now we introduce another important family of rings. the ideals.

They are not that unfamiliar,

For $a, b \in \mathbb{Z}$, the set $\{ax + by : x, y \in \mathbb{Z}\}$ is an ideal.

Def 1. Let R be a commutative ring.

A subset $I \subseteq R$ is an ideal if

$$(1) \quad 0 \in I.$$

$$(2) \quad \forall x, y \in I, \quad x - y \in I.$$

$$(3) \quad \forall r \in R, x \in I, \quad rx \in I.$$

eg. let $S \subseteq R$ be a subset

We define $\langle S \rangle = \bigcap_{S \subseteq I, I \text{ ideal}} I$.

This is the smallest ideal containing S .

We call it the ideal generated by set S .

In particular, $\langle S \rangle = \left\{ \sum_{i=1}^{n < \infty} r_i s_i : r_i \in R, s_i \in S \right\}$

eg. let $a \in R$. The set

$$\text{Ann}_R(a) = \{ r \in R : ra = 0 \}$$

is an ideal.

Lemma 2: Let R be a commutative ring.

and $I \subseteq R$ be an ideal. Then,

$$(1) \quad I = R \quad \text{iff} \quad \exists \text{ unit } u \in I.$$

$$(2) \quad R \text{ is a field iff } R \text{ has} \\ \text{only two ideals. } (0) \text{ and } R.$$

proof: (1). We note that

$$I = R \quad \text{iff} \quad 1 \in I \quad \text{iff} \quad u \in I.$$

(Convince yourself each step is true)

$$(2) \quad (\Rightarrow) \quad R \text{ being a field}$$

implies every nonzero element is a unit.

So any nonzero ideal contains at least one unit, which is R by (1).

($\Leftarrow$) Assume R has only two ideals

(0) and R . Let $0 \neq r \in R$

Then $(r) \neq (0)$ and hence $(r) = R$.

so $1 \in (r)$ and r is a unit.

Def 3. Let R be a commutative ring

and $I \subsetneq R$ a proper ideal. We say

(1) I is a prime ideal if

whenever $ab \in I$ for $a, b \in R$.

eg. Ideals are subgroups.

So ideals of $\mathbb{Z}$ is of the form (d) for some $d \in \mathbb{Z}$.

Then (d) is prime iff d is prime
and $d \neq 0$

We claim (d) is maximal iff
 d is prime. (why)

Def 6. Let R be a commutative

ring. We say R is Noetherian

if any ascending chain of
ideals eventually stabilizes.

That is, if $0 = I_0 \subseteq I_1 \subseteq I_2 \subseteq \dots$

then $\exists N \in \mathbb{N}$ s.t. $I_N = I_{N+k} \forall k \in \mathbb{N}$

eg, $\mathbb{Z}$ is Noetherian.

Let $0 = I_0 \subseteq I_1 \subseteq I_2 \subseteq \dots$

be an ascending chain of ideals

Write $0 = I_0 \subseteq (d_1) \subseteq (d_2) \subseteq \dots$

for $d_i \in \mathbb{Z}$.

Then $d_2 \mid d_1$, $d_3 \mid d_2$, $\dots$

in general. $d_{i+1} \mid d_i \quad \forall i \geq 1$

But d_1 has finitely many

divisors. so eventually d_i stabilizes

Or one may consider

$$d_1 \geq d_2 \geq d_3 \geq \dots$$

the descending chain of positive integers

admits a least element.